C++ Project on

# DEPARTMENT LIBRARY MANAGEMENT SYSTEM

## Submitted To:

U.VIJAYALAKSHMI MAM

EC3202-DATA STRUCTURES AND PROGRAMMING IN C++

## Submitted By:

GIRISURYAA L S

2023504022

B.E-ECE(1st YEAR)

# INTRODUCTION:

**Introduction To Department Library Management System:**

As the main objective of this C++ project on Department library Management System is to manage the books available in the library and students transactions as per date and their last visit and the students easily check their transactions history. The limit of the books is four for each
.

**Classes Used In This Program:**

1. class biodata

- It is the base class of this program.
- In this class there are
  void personal ()
  void name_check()
  void print()
  are the functions used.

2. class book

- It is the derived class which is inherited from class biodata.
- In this class there are
  void book_buy()
  void print_data()
  are the functions used.

3. class welcome

- It is the derived class which is inherited from class book.
- In this class there are
  void inpass()
  void pass_check()
  are the functions used.

**Functions Used In The Program:**

a. void personal()

This function is used for registration of students by given their details such as

(1) Name
(2) Register Number
(3) Year of study

When it saved, it shows the notification "Saved Successfully ✓ !!! "

b. void name_check()

This function is used to check their array number with using their register number as input. The program will run only they are registered , otherwise it shows warning. This function plays major role in identifying their role in this Department Library.

c. void print()

This function is used to print their name with a welcome address by using their register number.

d. void book_buy()

This function is used to buy the books for the students from the Library. The limit of the books is four for each student. This function also store their last visit to the Library. If they like to take more than four book, it doesn't allow to register the books.

e. void print_data()

This function is used for the students to see their book transactions (data) and their last visit to the Library. This is very useful to see their limits of the books📚.

f. void inpass()

This function is directly call by main function { main() } . This function is used to get password as input from the consumer of the Library and make a pathway to check the password using the function pass_check().

g. void pass_check()

This function plays very very important role in this program because it takes the password as argument and connect all the functions to play corresponding to the password.

# ABSTRACT:

The main objective of this C++ project on Department Library Management System is to manage the books in the Library and their buy and return from the students.

It manages all the information about the details of students ( Note: If they are registered using this program ) , their transactions and their last visit.

The project is built for the billing purpose at the customer level as it reduces the time for the billing management and the manual work for the managing items and orders.

It tracks all the details about books available and students transactions as well as the Library management.

# PROGRAM CODE:

```
#include<iostream>
#include<cstring>
using namespace std;

class biodata
{
  public:
        string name[50];
          int    reg[50] ;
          int    year[50];
        static int i;
     int    b[20]={0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0};
        int    ireg;
        int    k=-1;
     void personal()
    {
      cout<<"ENTER YOUR NAME         : ";
      cin>>name[i];
      cout<<"ENTER YOUR REGISTER NO. : ";
      cin>>reg[i] ;
      cout<<"ENTER YOUR YEAR OF STUDY: ";
      cin>>year[i];
      cout<<"Saved Successfully ✓ !!!\n\n";
      i++;
    }
     void name_check()
    {
      cout<<"Enter register number: ";
      cin>>ireg;
      for(int j=0;j<50;j++)
          {
       if(reg[j]==ireg)
      {
```

```
         k=j;
                  break;
            }
        else
       {
         k=-1;
       }
           }
         }
           void print()
         {
            cout<<"Welcome☺!!! "<<name[k]<<endl;
         }
};
int biodata::i=3;

class book : public biodata
{
  public:
      int    book_no[50][4];
      string c[50] ;
      int    ib;
      void book_buy()
     {
       cout<<"\nEnter the date : ";
       cin>>c[k];
       cout<<"\nHow many books do you want to buy? : ";
       cin>>ib;
       int p=ib+b[k];
       if(p<=4)
         {
             for(int j=b[k];j<p;j++)
            {
         cout<<"Enter book "<<j+1<<" number : ";
         cin>>book_no[k][j];
              b[k]++;
            }
```

```
            cout<<"Gain More Knowledge, Thank you🙏\n\n";
           }
            else
           {
             cout<<"You have able to take only "<<4-b[k]<<" books 📚\n\n ";

         }
      }
       void print_data()
      {
        cout<<"\nNumber of books 📚 you have is "<<b[k];
        for(int j=0;j<b[k];j++)
       {
         cout<<"\nBook "<<j+1<<" is "<<book_no[k][j];
       }
        if(b[k]!=0)
       {
         cout<<"\nYou last visited us on "<<c[k]<<endl<<endl;
       }
        else
       {
         cout<<"\nYou only register in this library\n\n";
       }
      }
};

class welcome : public book
{
  public:
         string pass;
         void inpass()
      {
            cin>>pass;
            pass_check(pass);
          }
         void pass_check(string pass)
         {
```

```
if(pass=="register")
{
personal();
}
else if(pass=="hi")
{
name_check();
if(k>-1)
{
print();
book_buy();
}
else
{
cout<<"Register your name first...☞\n\n";
}
}
else if(pass=="data")
{
name_check();
if(k>-1)
{
print();
print_data();
}
else
{
cout<<"Register your name first...☞\n\n";
}
}
else if(pass=="owner")
{
for(int j=0;j<i;j++)
{
cout<<"\n"<<name[j]<<"("<<reg[j]<<") - "
<<year[j]<<" year\n"<<b[j]<<" books - "
<<c[j]<<" last book\n";
```

```
        }
         cout<<"\nTill now "<<i<<" students visit our library\n\n ";
       }
        else if(pass=="about")
       {
         cout<<"As the main objective of this C++\nproject on Department
library Management\nSystem is to manage the books available\nin the
library and students transactions\nas per date and their last visit and
the\nstudents easily check their transactions\rhistory. The limit of the
books is four \nfor each.Thank you for search about me□\n\n";
       }
        else
       {
         cout<<"Incorrect password 👎\n\n";
       }
      }
};
 int main()
{
  welcome a;
  cout<<"WELCOME TO DEPARTMENT LIBRARY\n";
  cout<<"    (😎Girisuryaa's Code)   \n\n";
  a.name[0]="GIRISURYAA_L_S";
  a.reg[0]=2023504022;
  a.year[0]=1;
  a.name[1]="SAKTHIVEL_J";
  a.reg[1]=2023504035;
  a.year[1]=1;
  a.name[2]="HARIPRASATH_S";
  a.reg[2]=2023504030;
  a.year[2]=1;
  for(int i=0;i<100;i++)
 {
   a.inpass();
 }
  return 0;
}
```

# EXAMPLE OUTPUT:

```
WELCOME TO DEPARTMENT LIBRARY
          (Girisuryaa's Code)

register
ENTER YOUR NAME        : GIRISURYAA_L_S
ENTER YOUR REGISTER NO. : 2023504022
ENTER YOUR YEAR OF STUDY: 1
Saved Successfully …!!!

hi
Enter register number: 2023504022
Welcome…!!! GIRISURYAA_L_S

Enter the date : 23/04/2024

How many books do you want to buy? : 2
Enter book 1 number : 2324
Enter book 2 number : 3241
Gain More Knowledge, Thank you…!!!

data
Enter register number: 2023504022
Welcome…!!! GIRISURYAA_L_S

Number of books you have is 2
Book 1 is 2324
Book 2 is 3241
You last visited us on 23/04/2024
```

about
As the main objective of this C++
project on Department library Management
System is to manage the books available
in the library and students transactions
as per date and their last visit and the
history. The limit of the books is four
for each.Thank you for search about me.

owner

GIRISURYAA_L_S(2023504022) - 1 year
2 books - 23/04/2024 last book

SAKTHIVEL_J(2023504035) - 1 year
0 books -  last book

HARIPRASATH_S(2023504030) - 1 year
0 books -  last book

Till now 3 students visit our library

# WORKING:

This program contains 4 types of passwords to access Department Library books 📚.

(1) register

"register" is a password to register student's details in this program to access the books from the Department Library.

(2) hi

"hi" is a password to register the books details such as how many books want to take and the book's number by using their register number .

(3) data

"data" is a password to see their book transactions from the Library and it also shows their last visit by using their register number.

(4) owner

"owner" is a password to access all the student's transactions and their last visit to the library. It is a password for the Library Incharges to manage Department Library.

(5) about

"about" is a password to know about the main objective of this program for the Department Library Management System.

# CONCLUSION:

I performed this program in "CODING C++" application which is efficient in maintaining Students's details and can easily perform operations on students's records and also works to handles the information of the books that available with the students.

This software also reduces the work load of the Department Library incharges and They easily check the history of the students who bought books from Library.

There can be certain modifications such as edit the return books from the students.

***Thank You…!!!***